Hoigi Seo

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Education

- **Seoul National University** — Seoul, Republic of Korea
  B.E. in Electrical and Computer Engineering — *March 2018 - Feburary 2024*
- **Seoul National University** — Seoul, Republic of Korea
  Ph.D. in Electrical and Computer Engineering — *March 2024 - Present*

Research Interest

- **Multi-Modal Generative Models**: Multi-modal large language model, Text-to-image / video diffusion model
- **Efficient Model Training & Inference**: Quantized model training, Diffusion model acceleration

Publications (AI top-tier conference)

- **Triadic Dynamics Aware Diffusion Posterior Sampling for Inverse Problems: Optimizing Guidance and Stochasticity Schedules**
  J Bang*, D Moon*, **Hoigi Seo**, S Hong & S Y Chun. (*co-first authors)
  International Conference on Machine Learning (**ICML**), 2026.
- **Erasing Thousands of Concepts: Towards Scalable and Practical Concept Erasure for Text-to-Image Diffusion Models**
  Hoigi Seo*, B H Lee*, Jaehyun Cho, Sungjin Lim & S Y Chun. (*co-first authors)
  Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026.
- **Training free, Perceptually Consistent Low-Resolution Previews with High-Resolution Image for Efficient Workflows of Diffusion Models**
  W Jeong*, **Hoigi Seo*** & S Y Chun. (*co-first authors)
  Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026.
- **Training-free Mixed-Resolution Latent Upsampling for Spatially Accelerated Diffusion Transformers**
  W Jeong*, K Lee*, **Hoigi Seo** & S Y Chun. (*co-first authors)
  Conference on Computer Vision and Pattern Recognition (**CVPR**) **Highlight**, 2026.
- **On Epistemic Uncertainty of Visual Tokens for Object Hallucinations in Large Vision-Language Models**
  Hoigi Seo*, D U Kang*, H Cho, J Lee & S Y Chun. (*co-first authors)
  Neural Information Processing Systems (**NeurIPS**), 2025.
- **Skrr: Skip and Re-use Text Encoder Layers for Memory Efficient Text-to-Image Generation**
  Hoigi Seo*, W Jeong*, J Seo & S Y Chun. (*co-first authors)
  International Conference on Machine Learning (**ICML**), 2025.
- **Efficient Personalization of Quantized Diffusion Model without Backpropagation**
  Hoigi Seo*, W Jeong*, K Lee & S Y Chun. (*co-first authors)
  Conference on Computer Vision and Pattern Recognition (**CVPR**), 2025.
- **INTRA: Interaction Relationship-Aware Weakly Supervised Affordance Grounding**
  J H Jang*, **Hoigi Seo*** & S Y Chun. (*co-first authors)
  European Conference on Computer Vision (**ECCV**), 2024.
- **BeyondScene: Higher-Resolution Human-Centric Scene Generation with Pretrained Diffusion**
  G Kim*, H Kim*, **Hoigi Seo***, D U Kang* & S Y Chun. (*co-first authors)
  European Conference on Computer Vision (**ECCV**), 2024.

Preprints

- **Recovering Diversity in Distilled Diffusion Models via Low-Rank Policy-Optimized Merging**
  Hoigi Seo, W Han, J Lee & S Y Chun.
  ECCV Under review, 2026.
- **Geometrical Properties of Text Token Embeddings for Strong Semantic Binding in Text-to-Image Generation**
  Hoigi Seo*, J Bang*, H Lee*, J Lee, B H Lee & S Y Chun. (*co-first authors)
  arXiv preprint, 2025.
- **DITTO-NeRF: Diffusion-based Iterative Text To Omni-directional 3D Model**
  Hoigi Seo*, H Kim*, G Kim* & S Y Chun. (*co-first authors)
  arXiv preprint, 2023.

Skills Summary

- **Language** English (professional), Korean (native)
- **Programming Languages** Python, C, C++, Bash
- **Deep Learning Framework** PyTorch, TensorFlow

Patents

- **Training-free Industrial Defect Segmentation.**
  S. Y. Chun, D. U. Kang, H. Kim, J. H. Jang, B. H. Lee & **Hoigi Seo**
  Korean Patent, Filed. 2025.
- **Device and Method For Affordance Grounding.**
  S. Y. Chun, J. H. Jang & **Hoigi Seo**.
  Korean Patent, Filed. No. 10-2025-0029227, 2025.
- **Device and Method for Extracting High-Dimensional Gene Features.**
  S. Y. Chun, **Hoigi Seo**, H. Bae, D. U. Kang & H. Kim.
  Korean Patent, Filed. No. 10-2025-0014990, 2025.
- **Higher-Resolution Human-Centric Scene Generation With Pretrained Diffusion.**
  S. Y. Chun, G. Kim, H. Kim, **Hoigi Seo** & D. U. Kang.
  Korean Patent, Filed, No. 10-2024-0115484, 2024 / U.S. Patent Application, Filed, No. 19/019,060, 2025.

Honors and Awards

- Excellent Research Talent Fellowship (Korea Research Foundation) — September, 2025
- Brain Korea 21 Scholarship (Korea Research Foundation) — March, 2024 - Present
- Academic Excellence Scholarship (Seoul National University, ECE) — September, 2023

Projects

- **Megabase-scale DNA foundation model** — **Project Lead**
  NVIDIA Academic Grant (32K A100 hours) — *April 2026 - Present*
  - Training foundation models for Megabase-scale DNA foundation model for structural variant detection.
- **ecDNA Detection** — **Project Lead**
  Ministry of Science and ICT, Republic of Korea — *September 2023 - Present*
  - Extra-chromosomal DNA (ecDNA) detection and associated gene discovery from FISH images and gene profiles.
- **AI for Human-Machine Teaming** — Project Participants
  United States Air Force & IITP, Republic of Korea — *March 2025 - Present*
  - Development of human-robot interaction (HRI) and decision-making that can enable robots to collaborate naturally with humans with hazy oracle.
- **Efficient Personalization** — Project Participants
  Samsung Mobile Experience (MX) — *March 2024 - May 2025*
  - Memory and time efficient text-to-image synthesis and personalization for on-device deployment.
- **Industrial Defect Segmentation** — Project Participants
  Samsung Device Solution (DS) — *March 2024 - March 2025*
  - Development of human-robot interaction (HRI) and decision-making that can enable robots to collaborate naturally with humans with hazy oracle.

Other Publications

- **Tunable Porous Collagen Hydrogels as a Physiologically Relevant Platform for Extrachromosomal DNA-Associated Colorectal Cancer Research**
  S Jo, J Shon, S An, Y Nam, D Choi, S Lee, **Hoigi Seo**, S Y Chun & H Kim.
  Theranotics (IF: 13.3), 2025.
- **ecSegCls: Deep learning-based method for detecting extrachromosomal DNAs in both interphase and metaphase cancer cells**
  Hoigi Seo, Y Nam, H Bae, D U Kang, J Lee, H Kim & S Y Chun.
  American Association for Cancer Research (**AACR**), 2026.
- **Harmonization for a black-box deep learning model**
  M Kim, H Jeong, **Hoigi Seo**, W Jeong, S Y Chun & J Lee.
  International Society for Magnetic Resonance in Medicine (**ISMRM**) **(Oral, Summa Cum Laude, AMPC select)**, 2025
- **Mitigating Hallucinations in Large Vision-Language Models via Norm-Guided Visual Filtering**
  H Cho, **Hoigi Seo**, D U Kang, J Lee & S Y Chun.
  Conference on Electronics, Semiconductor and AI, 2025.
- **Towards Personalization of Generative Models Via Zeroth-Order Optimization**
  W Jeong, **Hoigi Seo** & S Y Chun.
  Korea Signal Processing Conference, 2024.

Other Experience

- **Military Service** — Sergeant
  Served as a Situation Room Soldier in the 3rd Guard Battalion. — *May 2020 - November 2021*